

VIDEO COURSE

Video Course: DocProcessor Deep Dive

5-episode series covering DocProcessor from fundamentals to advanced usage.

Episode 1: Introduction & Architecture (15 min)

Script Outline

1. Opening (2 min)

- What is DocProcessor and why it exists
- Role in the HelixQA ecosystem
- Problem statement: manual QA doesn't scale

2. Architecture Overview (5 min)

- Package layout: loader, feature, coverage, docgraph, llm, config
- Processing pipeline visualization
- Thread safety design with RWMutex

3. Quick Start Demo (5 min)

- Clone and build
- Run CLI on sample docs
- Show feature map output

4. Key Design Decisions (3 min)

- LLMAgent injection (no hard dependencies)
 - Deterministic feature IDs
 - Heuristic fallback when no LLM
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Episode 2: Document Loading & Parsing (15 min)

Script Outline

1. Loader Interface (3 min)

- LoadDir, LoadFile, SupportedFormats
- Context cancellation support
- MaxFileSize protection

2. Markdown Parser (4 min)

- Heading extraction with regex
- Section content segmentation
- Link extraction and deduplication

3. YAML Parser (3 min)

- Generic unmarshalling
- Title extraction from title or name keys
- Section creation from top-level keys

4. Scanner Implementation (3 min)

- Recursive directory walking
- Hidden directory skipping
- Extension normalization (yaml -> yml)

5. Security (2 min)

- Path traversal protection
 - File size limits
 - Malformed input handling
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Episode 3: Feature Extraction & FeatureMap (15 min)

Script Outline

1. Feature Model (3 min)

- Feature struct: ID, Name, Category, Platforms, Priority
- FeatureCategory constants
- Deterministic ID generation

2. **Heuristic Extraction** (3 min)

- Section-based extraction
- Content length thresholds
- Default platform assignment

3. **LLM-Powered Extraction** (4 min)

- LLMAgent interface design
- Prompt templates with versioning
- ExtractFeatures, ClassifyFeature flow

4. **FeatureMap Building** (3 min)

- AddFeature with category/platform indexing
- DocGraph integration
- FeatureByID, FeaturesForPlatform queries

5. **Merge & Deduplication** (2 min)

- Merging multiple feature maps
 - ID-based deduplication
 - DocGraph merging
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Episode 4: Coverage Tracking (15 min)

Script Outline

1. **CoverageTracker Interface** (3 min)

- RegisterFeature, MarkVerified, MarkFailed, MarkSkipped
- Coverage reports by platform and category
- Export for serialization

2. **Thread-Safe Implementation** (4 min)

- sync.RWMutex usage
- Read path: RLock for Coverage(), CoverageByPlatform(), Unverified()
- Write path: Lock for MarkVerified(), MarkFailed(), MarkSkipped()

3. **Verification States** (3 min)

- State machine: unverified -> verified/failed/skipped
- State transitions and overwrites
- Evidence and Issue tracking

4. **Reporting** (3 min)

- CoverageReport aggregation

- Platform and category breakdowns
- Issue collection

5. **Stress Testing** (2 min)

- Concurrent verification from multiple platform workers
 - Race condition detection
 - High-contention scenarios
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Episode 5: DocGraph & Integration (15 min)

Script Outline

1. **DocGraph** (4 min)

- Directed graph of document links
- Node and edge management
- Neighbor and incoming edge queries

2. **Export Formats** (3 min)

- JSON export and import (round-trip)
- Mermaid diagram generation
- ID sanitization for Mermaid

3. **Full Pipeline Integration** (4 min)

- Load -> Parse -> Extract -> Build -> Enrich -> Track
- Mock LLMAgent for testing
- E2E test walkthrough

4. **Configuration** (2 min)

- .env file loading
- Default values
- Environment variable mapping

5. **What's Next** (2 min)

- HelixQA integration via bridge adapters
- LLMArchestrator Agent -> LLMAgent adapter
- Autonomous QA session pipeline